Semester:	III	**Course Type:**	ETC		
Course Title: Python for Data Analytics					
Course Code:	**23CSE312**			**Credits:**	3
Teaching Hours/Week (L:T:P:O)			2:0:2:0	**Total Hours:**	25 +(10-12 lab slots)
CIE Marks:	50	**SEE Marks:**	50	**Total Marks:**	100
SEE Type:	Theory			**Exam Hours:**	3

I. Course Objectives:

This course will enable students to:

- Introduce fundamental Python programming concepts
- Develop proficiency with essential data structures
- Apply the NumPy library to perform mathematical operations
- Utilize Pandas DataFrames to explore and analyze data
- Employ Pandas functionalities to clean and prepare data for analysis

II. Pre-requisites: Basics of C and Python

III. Teaching-Learning Process (General Instructions):

These are sample Strategies, which teachers can use to accelerate the attainment of the various course outcomes.

1. Lecturer method (L) need not to be only a traditional lecture method, but alternative effective teaching methods could be adopted to attain the outcomes.

2. Use of Video/Animation to explain functioning of various concepts.

3. Encourage collaborative (Group Learning) Learning in the class.

4. Ask at least three HOT (Higher order Thinking) questions in the class, which promotes critical thinking.

5. Adopt Problem Based Learning (PBL), which fosters student's Analytical skills, develop design thinking skills such as the ability to design, evaluate, generalize, and analyze information rather than simply recall it.

6. Show the different ways to solve the same problem and encourage the students to come up with their own creative ways to solve them.

7. Discuss how every concept can be applied to the real world and when that's possible, it helps to improve the student's understanding.

IV. COURSE CONTENT

IV(a). Theory PART

Module-1: An Introduction to Data Analysis	Hrs 5
Data Analysis, Knowledge Domains of the Data Analyst, Understanding the Nature of the Data, The Data Analysis Process, Quantitative and Qualitative Data Analysis, Open Data, Python and Data Analysis **Text book 1: Chapter 1**	
RBT Levels: 1,2,3	
Module-2: Essentials of python programming	Hrs 5
Introduction to Python: Features of Python, Installation of Python,Variables in Python, Output in Python, Input in Python, Operators **Control Flow Statements:** Decision making structures, Loops, Abnormal Loop Termination, User-defined Functions **Data Structures:** Lists, Tuples, Dictionary **Text Book 2: Chapter 1: 1.1, 1.2, 1.4, 1.5, 1.6, 1.7, Chapter 2: 2.1, 2.2, 2.4, 2.6, Chapter 3: 3.1, 3.2, 3.3**	
RBT Levels: 1,2,3	
Module-3: Introduction to Numpy library	Hrs 5
The NumPy Library: ndarray: the heart of the library, Basic operations, indexing, slicing and iterating, conditions and boolean arrays, array manipulation, general concepts, reading and writing array data on files. **Textbook 1: Specified topics from Chapter 3**	
RBT Levels:1,2,3	
Module-4: Introduction to Pandas library	Hrs 5
The pandas Library: An Introduction to Data structure, Other functionalities on indexes, Operations between Data Structures, Function Application and Mapping. Reading & Writing Data: I/O API tools, csv & Textual Files, Reading data in CSV or text files, Reading and Writing HTML files, Reading & Writing Data on Microsoft Excel Files. **Textbook 1: Specified topics from Chapter 4 and 5**	
RBT Levels:2,3,4	
Module-5: Data Visualization with matplotlib	Hrs 5
The matplotlib Library, matplotlib Architecture, pyplot, using the kwargs, Adding further elements to the chart, Saving charts, Handling data values, Line chart, Histogram, Bar chart, **Textbook 1: Specified topics from Chapter 7**	
RBT Levels:2,3	

IV(b). PRACTICAL PART

Sl. No.	Experiments / Programs / Problems
1	a. Write a Python program to determine whether a given number is a prime number or not. Also enhance the program to find the first N prime numbers, where N is user input. b. Write a Python program to accept a positive integer and determine if it is a perfect square. Also display a message accordingly.
2	Create a list to represent a shopping cart, containing items (strings). a. Add items to the cart (using append()) b. Remove an item from the cart (using remove()) c. Create a function add_multiple_items that takes a list of items and uses extend to add them to the shopping cart in one go. d. Implement a function capitalize_first_letter that modifies an item's name (capitalizing the first letter) before adding it to the cart.
3	Write a Python program that functions as a customizable text analyzer. The program should accept a block of text and provide options to choose the type of analysis(word count, character count, Uppercase count and list, Lowercase count and list). Define separate functions for each option and also display the result in user-friendly format.
4	Create a tuple containing your name, age, and favorite color. Print the elements individually and access the entire tuple.
5	Write a Python program using NumPy that creates a random m x n integer array and prints the shape of the array (number of rows and columns) and also total number of elements in the array. a. Implement function fill_with_value (e.g., zeros) b. Implement a function randomly_generate_numbers (within a range) to populate the array.
6	Develop a python program to read and print CSV file in the console. After reading the CSV, calculate summary statistics for numerical columns (mean, median, standard deviation) using pandas functions.
7	Write a Python program to demonstrate how to draw a bar plot using Matplotlib.
8	Write a Python program to demonstrate how to draw a histogram plot using Matplotlib.

V. COURSE OUTCOMES

CO1	Develop programs that demonstrate understanding of core concepts of Python language like data types, control flow, and functions.
CO2	Structure and organize data using lists, tuples and dictionaries in Python programs.
CO3	Apply essential numerical computations and data manipulation using the NumPy library.
CO4	Construct and manage DataFrames in pandas for data analysis tasks.
CO5	Implement basic plotting techniques using matplotlib library.

VI. CO-PO-PSO MAPPING (mark H=3; M=2; L=1)

PO/PSO	1	2	3	4	5	6	7	8	9	10	11	12	S1	S2	S3	S4
CO1	2	1	3		2									1		
CO2	2	1			1									1		
CO3	2	1	2		2								1	1		
CO4	2	1			1								1	1		
CO5	2	1			1								1	1		

VII. Assessment Details (CIE & SEE)

General Rules: Refer CIE and SEE guidelines based on course type for autonomous scheme 2023.

Continuous Internal Evaluation (CIE): Refer Annexure section 1

Semester End Examination (SEE): Refer Annexure section 1

VIII. Learning Resources

VIII(a): Textbooks: (Insert or delete rows as per requirement)

Sl. No.	Title of the Book	Name of the author	Edition and Year	Name of the publisher
1	**Python Data Analytics**	**Fabio Nelli**	**First Edition, 2015**	**Apress Publishing**
2	**Data Analytics using Python**	**Bharti Motwani**	**First Edition: 2020**	**Wiley Publication**

VIII(b): Reference Books: (Insert or delete rows as per requirement)

1	**Think Python**	**Allen B. Downey**	**Second Edition, Dec 2015**	**O'Reilly**
2	**Python for Data Analysis**	**Wes McKinney**	**Third Edition**	**O'Reilly**

VIII(c): Web links and Video Lectures (e-Resources):

WebLinks:

1. https://docs.python.org/3/tutorial/ (Official Python Tutorial)
2. https://www.learnpython.org/ (Learn Python website with interactive tutorials)
3. https://discuss.python.org/c/core-dev/23 (No Starch Press' Intro to Python for Beginners)
4. https://numpy.org/doc/ (NumPy Official Documentation)
5. https://pandas.pydata.org/docs/ (Pandas PyData Documentation)
6. https://realpython.com/lessons/len-numpy-and-pandas/ (How to Use NumPy and Pandas Together by Real Python)

Video Lectures:

1. https://www.youtube.com/watch?v=eAoZjemZXrM (Crash Course in Python for Data Science by Google)
2. https://www.youtube.com/watch?v=FniLzpaSFGk (NumPy & Pandas Essential Training for Python by Linda Sandvik on YouTube)
3. https://m.youtube.com/watch?v=VK2QfDXPg6k (Data Wrangling with Pandas by Real Python)
4. https://www.youtube.com/channel/UC8butISFwT-Wl7EV0hUK0BQ/search?query=Python (Data Cleaning in Python Tutorial by freeCodeCamp on YouTube)

IX: Activity Based Learning / Practical Based Learning/Experiential learning:

- Quizzes
- Assignments

Semester:	III	**Course Type:**	ETC		
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CIE Marks:	50	**SEE Marks:**	50	**Total Marks:**	100
SEE Type:	Theory			**Exam Hours:**	3

I. Course Objectives:

This course will enable students to:

- Understand fundamentals of ethical hacking, penetration testing, and the role of Linux in cybersecurity.
- Equip with essential Linux commands for text manipulation and basic network analysis techniques.
- Develop understanding of file and directory permissions, user management, and process control in Linux.
- Understand Bash scripting for automating tasks and provide them with tools for detecting network vulnerabilities.
- Familiarize with the Metasploit Framework, a popular tool for penetration testing and exploiting vulnerabilities.

II. Teaching-Learning Process (General Instructions):

These are sample Strategies, which teachers can use to accelerate the attainment of the various course outcomes.

1. Lecturer method (L) need not to be only a traditional lecture method, but alternative effective teaching methods could be adopted to attain the outcomes.
2. Use of Video/Animation to explain functioning of various concepts.
3. Encourage collaborative (Group Learning) Learning in the class.
4. Ask at least three HOT (Higher order Thinking) questions in the class, which promotes critical thinking.
5. Adopt Problem Based Learning (PBL), which fosters student's Analytical skills, develop design thinking skills such as the ability to design, evaluate, generalize, and analyze information rather than simply recall it.
6. Introduce Topics in manifold representations.
7. Show the different ways to solve the same problem and encourage the students to come up with their own creative ways to solve them.